Integrated Data Acquisition System using UART, SPI & I2C Protocol

PROJECT MEMBERS:

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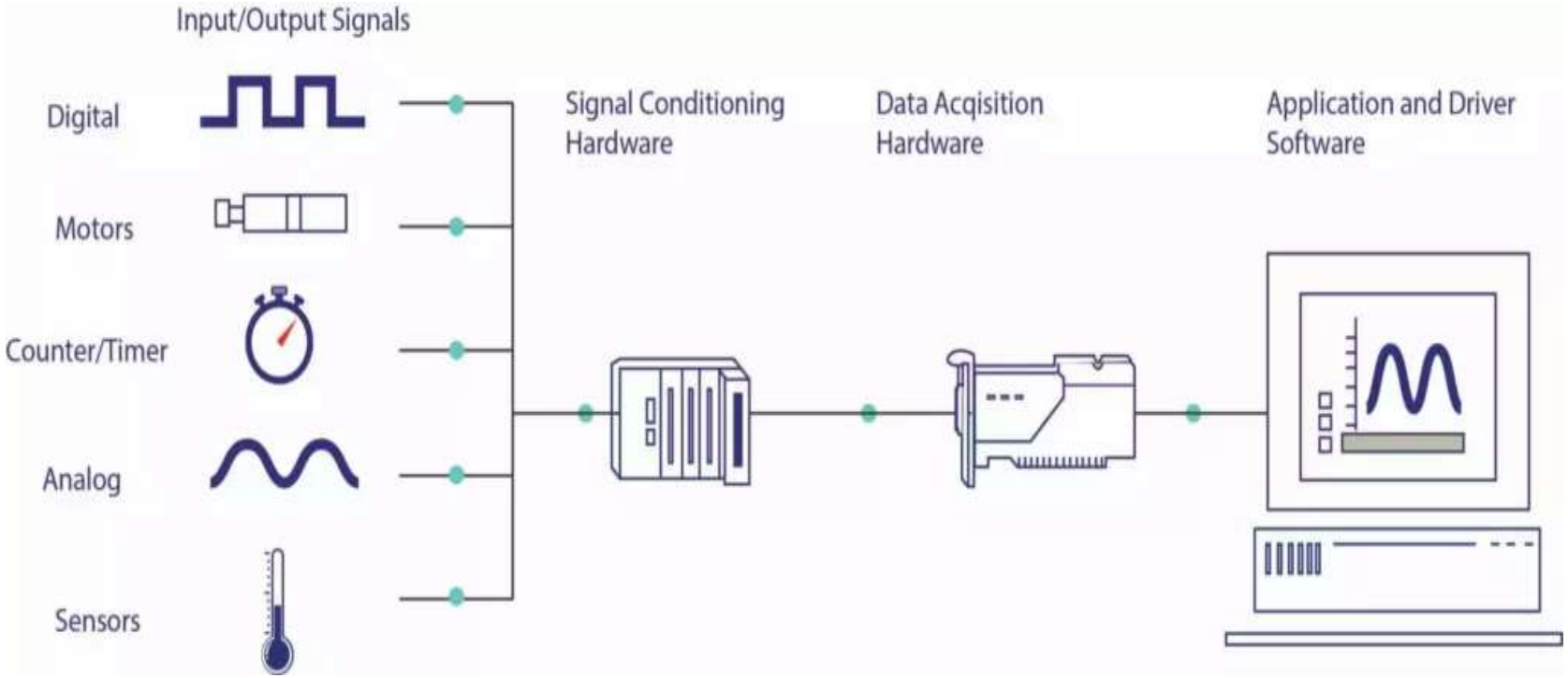


CONTENTS

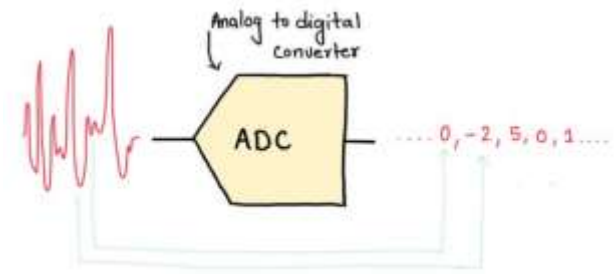
- Introduction
- Objective
- Proposed System
- Hardware/Software Used
- Implementation
- Results

INTRODUCTION

- Integrated Data Acquisition System (IDAS) is used to monitor key vehicle parameters in real time.
- Collects data from multiple sensors: temperature (LM35), light intensity (LDR), and voltage (POT).
- Uses LPC2129 microcontroller to process and display sensor values.
- RTC (DS1307) provides accurate real-time clock using the I2C protocol.
- External ADC (MCP3204) reads LDR through the SPI protocol.
- Sensor data is displayed on a 16×2 LCD and transmitted to PC via UART.
- System helps in monitoring, analysis, and safety in vehicle environments.

OBJECTIVES

1. Collect real-time sensor data such as temperature, light intensity, and voltage.

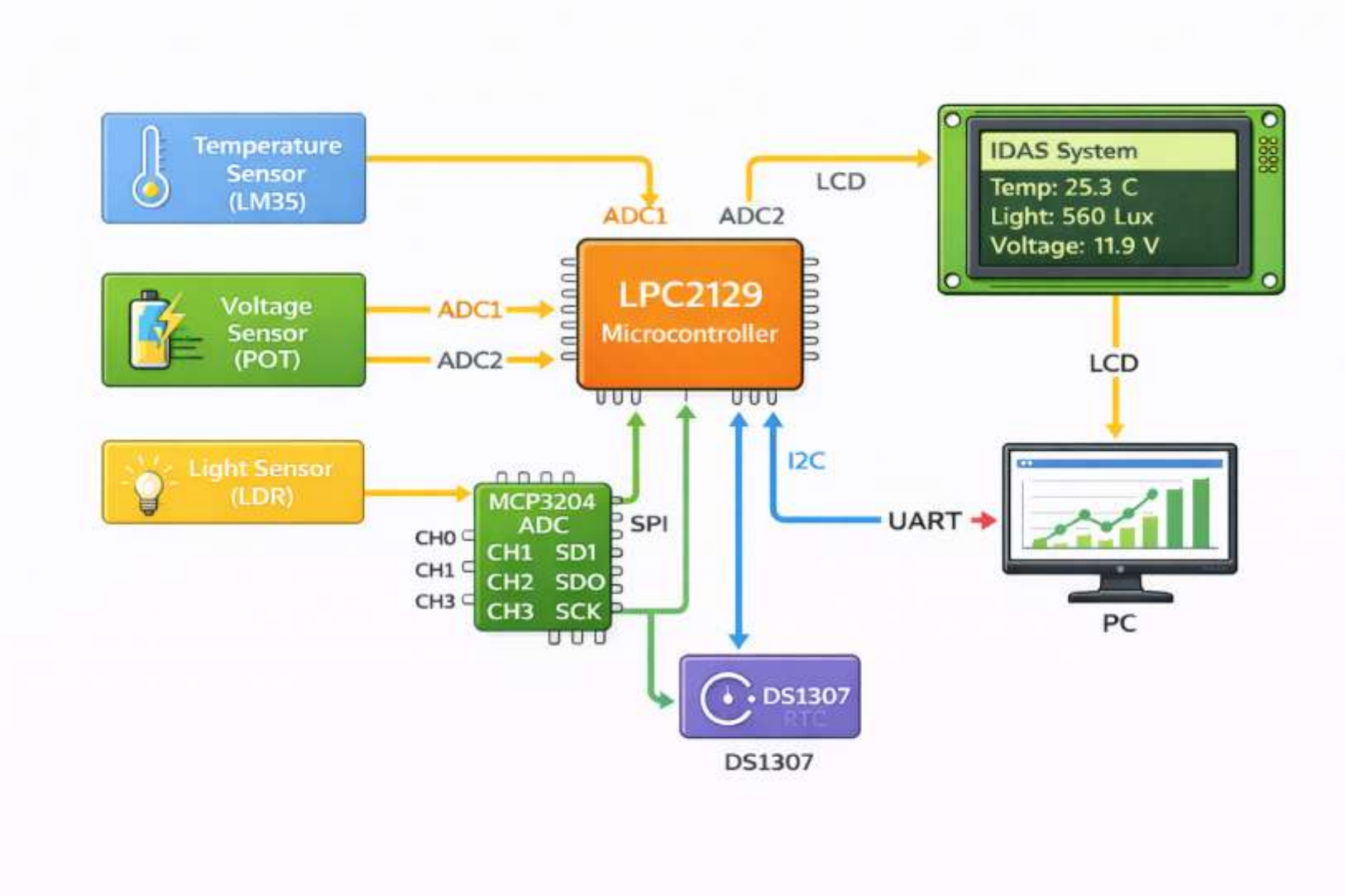


2. Ensure accurate measurement using on-chip ADC and external SPI-based ADC.

3. Display and transmit live data using LCD and UART communication.

4. Maintain precise timestamping using RTC (I2C) for monitoring and analysis.

PROPOSED BLOCK DIAGRAM



HARDWARE / SOFTWARE USED

❖ HARDWARE

1. LPC2129 Microcontroller

→ *The core processing unit of the system.*

2. External ADC (MCP3204)

→ *Converts analog sensor data into digital signals.*

3. Temperature Sensor (MCP9700)

→ *Measures ambient temperature accurately.*

4. RTC (DS1307)

→ *Maintains real-time clock and date information.*

5. LDR (Light Dependent Resistor)

→ *Detects surrounding light intensity.*

6. Power Supply Unit

→ *Provides stable operating voltage to the system.*

❖ SOFTWARE

1. Keil uVision IDE

→ *Used for coding, compiling, and debugging.*

3. Proteus

→ *Simulation tool for circuit testing.*

4. Flash Magic

→ *Uploads the HEX file into the microcontroller.*

COMPONENTS

1. External ADC

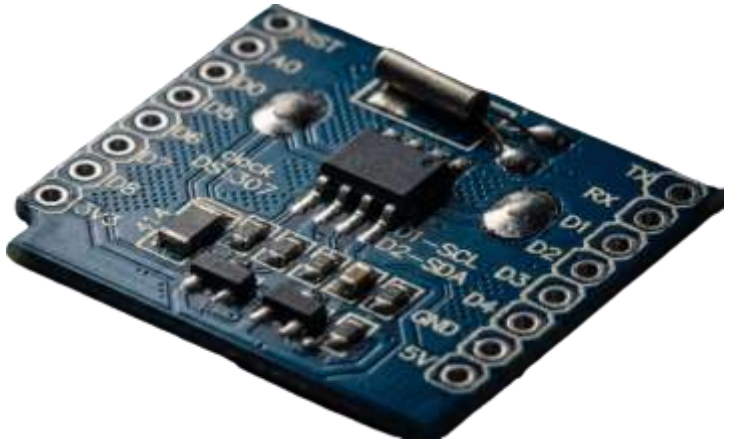
2. Rhydolabz board

3. LM53 Temp. Sensor

4. LDR Sensor

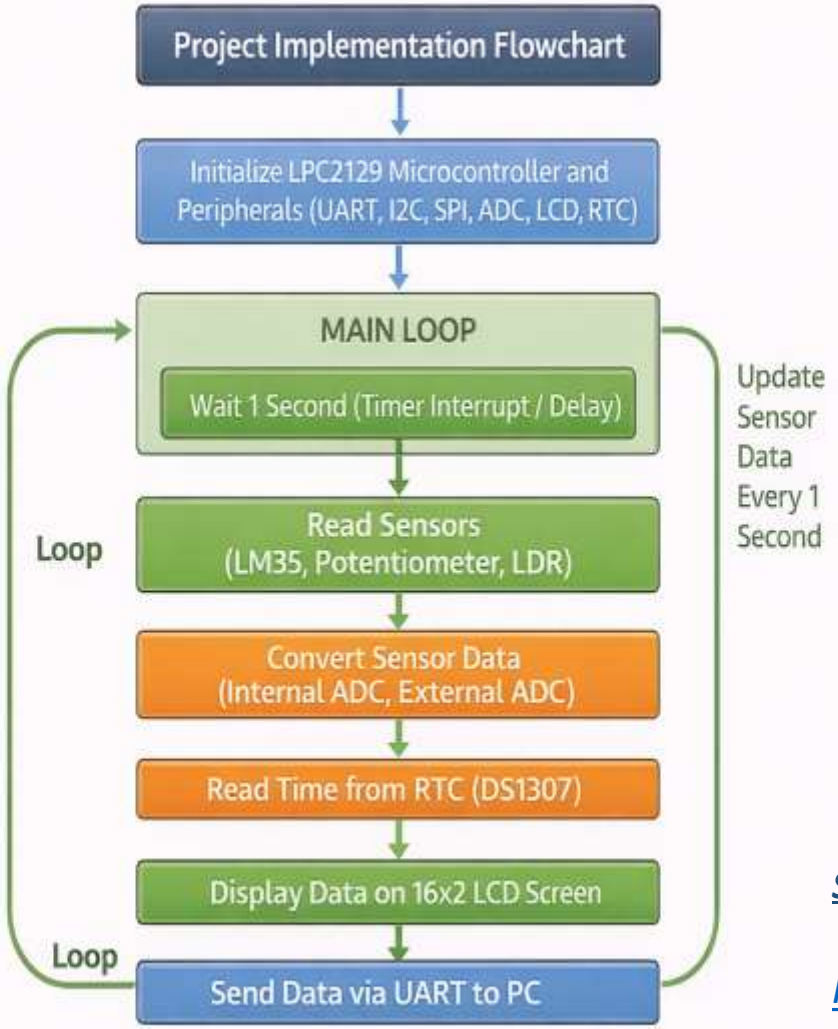
5. MCP3204

6. Potentiometer



7. DS 1307 RTC

IMPLEMENTATION



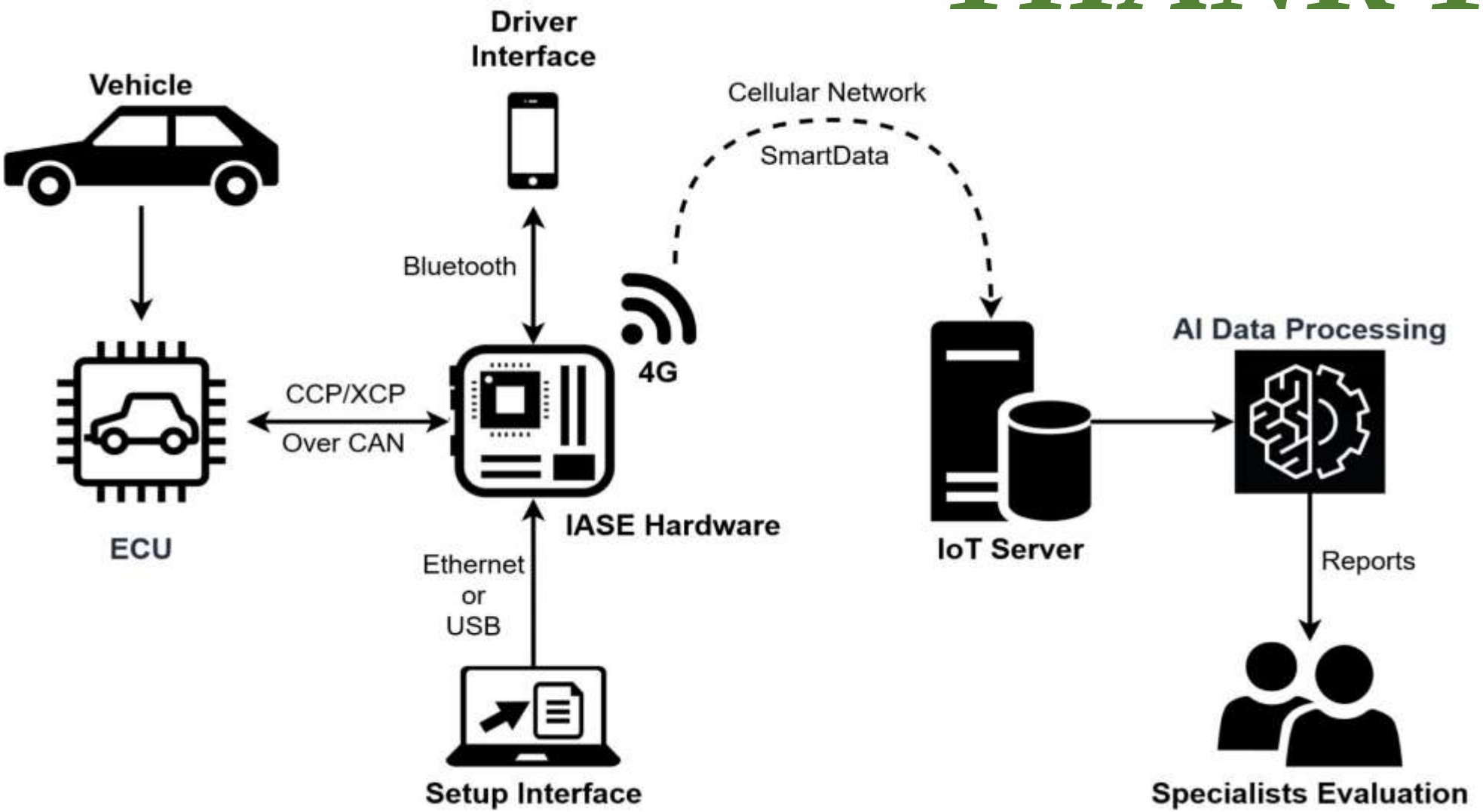
Simulation Video Link **:**

https://drive.google.com/file/d/1-DMqWUupzn9P5SFxmNqxZPQbTCyyEyOl/view?usp=sharing

RESULT

- Successfully implemented an Integrated Data Acquisition System (IDAS) using LPC2129.
- Real-time monitoring of Temperature, Light and Voltage was achieved.
- Sensor values were accurately converted using Internal ADC & External SPI ADC (MCP3204).
- The current time and date were displayed using the DS1307 RTC module via I2C.
- Processed data was displayed on 16x2 LCD and also transmitted to PC over UART.
- The system updates data every 1 second, ensuring continuous monitoring.
- The developed system performed reliably and efficiently during testing.

THANK YOU


Driver
Interface
Vehicle
Cellular Network
SmartData
Bluetooth
4G
AI Data Processing
CCP/XCP
Over CAN
IASE Hardware
ECU
IoT Server
Reports
Ethernet
or
USB
Setup Interface
Specialists Evaluation